Reproducing the redshifty Approach-A Redshift-Ignition Result on Commodity GPUs

Greg Benson Jonathan Samuel Xiaosheng Huang

June 2026
`commit: 6ef37db`

Phase 13 Internal Reproduction Report

Abstract

We reproduce the Approach-A "redshift ignition" result reported in the `cosmologyfoundation/redshifty` Phase-10 final NERSC run (Samuel, 2026) on a single non-NERSC commodity GPU (NVIDIA L4/A16 class). On a 4-way DESI EDR/DR1 data mix (sv3+main × bright+dark, 1.82M raw spectra), the local Approach-A model crosses the sustained `val_redshift_acc` $\geq$ 10% threshold by step 9000, with `val_redshift_acc` peak 14.86% at step 9500 and a final `val_loss` of 190.67 — the lowest of any of our five comparison arms. The `ar_redshift_acc`/`val_redshift_acc` ratio of 0.73 at step 9000 closely matches the NERSC reference value of 0.74 (`val_redshift_acc` = 73.8%, AR = 55.0%). A four-hypothesis diagnostic ladder shows that the sv3+main × bright+dark data mix is the load-bearing requirement: hyperparameter matching, data scale, and random-seed luck were all individually insufficient.

Contents

1 Introduction

This report documents Phase 13 of the agentic-lensing reproductions program. Phase 13 differs from prior phases in that the target is not a published external paper but an internal SpectrumFM-precursor project: the `redshifty` Approach-A transformer (Samuel, 2026), a unimodal DESI-spectra foundation model originally a USF PHYS303 final project. The reproduction question is whether the published Phase-10 NERSC training-curve result — a sharp "ignition" transition during which the encoder learns to embed redshift into its representation, raising `val_redshift_acc` from the noise floor (∼1–5%) to the 73.8% teacher-forced regime — is reproducible on the commodity-GPU hardware that the SpectrumFM agentic-AI tooling track will run on as Phase II compute scales out.

The motivation is twofold. First, validating the Phase-10 NERSC trajectory under our hands rules out NERSC-specific machinery (I/O, CFS, ERCAP allocations) as a confound on the architecture claim in the SpectrumFM Phase-I proposal: "one model, six classes". Second, producing a credible local Approach-A baseline gives us a head-to-head target to test the `cosmologyfoundation/codecs` Mamba3+RFSQ tokenizer against, which is the Phase-I architectural claim that needs to be validated in months 1–3 per the proposal milestones.

This report follows the structure of the prior reproduction tech reports under `reproductions/`. The work it documents lives in three places in the host `agentic-lensing` repository: `tools/spectrumfm/` (the experiment-running harness), `experiments/specs/` and `experiments/runs/` (per-run artifacts), and the supporting data tree at `/raid/benson/data/desi_dr1_medium/` (out of repo, ∼757 GiB).

2 Local toolchain

A small two-CLI harness drives all the experiments through a uniform yaml-spec interface (`tools/spectrumfm/exp_r` and `tools/spectrumfm/exp_analyze.py`). The harness:

- Reads an `experiments/specs/*.yaml` spec, resolves the target Python virtual environment (`redshifty` or `codecs`) from the spec's `repo` field, and launches the training subprocess.
- Tees the child's stdout to a per-run log file while parsing the `[step N] k=v k=v ...` and `[val N] k=v ...` metric lines emitted by `nersc/pretrain_tokenizer.py` and `nersc/train_transformer.py`, appending one JSON line per val record to a structured `metrics.jsonl`.
- On completion, writes a markdown `summary.md` (run configuration, headline statistics, command, wallclock) in the style of the `redshifty/RESEARCH_LOG.md` sections, so every run is auditable from the harness output alone.

Data acquisition is handled by a separate utility, `tools/spectrumfm/download_desi_subset.py`. It speaks the DESI public data-server URL conventions for both the EDR (`fuji`) and DR1 (`iron`) productions, supports a `--release {edr,dr1}` flag, is resumable across interrupted pulls, and arranges the downloaded coadd and Redrock files into the on-disk healpix tree shape that `redshifty/nersc/build_dr1_index.py` expects.

Total tooling code added for this work: ~ 350 lines of Python across the three CLIs. The harness has otherwise been used unchanged to drive all five experimental arms reported here.

3 V1 tokenizer reproduction

Approach-A in Samuel (2026) consumes discrete spectrum tokens from a frozen pretrained tokenizer (the "V1" tokenizer: ConvNeXt-V2 encoder/decoder (Woo et al., 2023) with Look-Up-Free Quantization (Yu et al., 2024)). Before reproducing Approach A we have to first reproduce the V1 tokenizer to a comparable `val_recon`.

We retrain V1 on 154 healpix files from the sv3-bright EDR/fuji subset (218k spectra) over 15000 steps with batch size 16 and base LR 3×10^{-4}, AMP enabled, on one NVIDIA L4. The trajectory descends monotonically with the only training-loop instabilities being two outlier-batch spikes that recover within the next val. The best-val checkpoint at step 13000 has `val_total` = 1.70 and `val_recon` = 1.38, vs. the published NERSC V1 baseline of 1.35 at step 16500 on 200 healpix / 394k spectra (Samuel 2026 Phase 8). We match the NERSC V1 result within 0.03 on `val_recon` using approximately half the training corpus.

We use this local V1 tokenizer (frozen) for every Approach-A run that follows. We did not pursue a V2 (U-Net + cross-attention) tokenizer locally; the published NERSC V2 result `val_recon` = 0.157 (Samuel, 2026) is dramatically lower than V1's 1.35, but the V1 tokenizer is the comparable baseline for the Phase-10 Approach-A run we are reproducing, and our spec/redshift accuracy results below are not bottlenecked by the tokenizer.

4 Approach-A diagnostic ladder

Our initial Approach-A run on 219k sv3-bright spectra produced `val_redshift_acc` peak 3.9% over 10,000 steps, with no sharp inflection and AR redshift accuracy never above the noise floor — nowhere close to the published Phase-10 figure of 73.8% (Samuel 2026 Phase 10). We tested four hypotheses for the gap, in sequence.

4.1 Hparam mismatch (necessary but not sufficient)

The initial spec accidentally combined Phase 9's `batch_size`=8, `lr`=2×10^{-4} (which Samuel (2026) used only with `encoder_mask_ratio`=0) with Phase 10's `encoder_mask_ratio`=0.50 — a combination the author never ran. Phase 10 final used `batch_size`=32, `lr`=4×10^{-4} (sqrt-scaled for the 4$\times$ batch bump). Re-running with the matched batch/LR pair raised peak `val_redshift_acc` from 3.9% to 8.2% on the same 219k spectra and dropped `val_loss` min from 224.7 to 199.2. Real and substantial improvement, but still nowhere near the 73.8% target.

4.2 Data scale (ruled out)

We hypothesized the gap was a data-quantity gap: 219k spectra vs. the author's 394k. We pulled the full sv3-bright tree (373 pixels / 729k spectra, 1.85$\times$ the author's 394k) and reran with the Phase 10 hparams. The trajectory matched the 219k-spectra run almost exactly — `val_redshift_acc` peak

6.18%, `val_loss` min 203.3, with the spectrum-side metrics improving (peak `val_spec_acc` = 45.4%, above the published NERSC ∼30%) but the redshift side flat. The data-quantity hypothesis is ruled out.

4.3 Random-seed luck (ruled out)

The cross-attention pathway through which the decoder reads redshift out of the encoder representation is a particular attention head's discovery, which is in principle initialization-sensitive. We ran a four-seed sweep on the xlarge data (seeds $\{42, 1337, 7, 12345\}$) with otherwise identical configuration. Per-seed peak `val_redshift_acc`: $6.18\%, 8.76\%, 3.76\%, 6.84\%$. The 8.76% on seed 1337 was a single-val noise spike (regressed to 1.33% next val); per-step variance across seeds was 1–3 pp standard deviation, dwarfed by the ∼70 pp gap to the published reference. Random-seed luck is ruled out.

4.4 Data mix (confirmed)

The Phase-10 setup line in the published RESEARCH_LOG.md reads "200 healpix files (sv3+main, bright+dark), 393967 spectra in flat index" — meaning the 200 pixels were divided across all four combinations of survey × program: ∼50 each of sv3-bright, sv3-dark, main-bright, main-dark. Our four prior arms used sv3-bright only, which is the most-common-on-paper interpretation of that line but is incorrect. We pulled the missing three combinations (EDR-fuji sv3-dark via the existing route; DR1-iron main-bright and main-dark via the `--release dr1` extension to `download_desi_subset.py`). The combined manifest has 1,137 pixels and 1.82M raw spectra (480k after the quality cut on `ZCAT_PRIMARY` ∧ `COADD_FIBERSTATUS`=0 ∧ `OBJTYPE`='TGT').

5 Ignition result on the mix data

Re-running the Phase-10 hparams on the mix manifest produced the result. Table 1 shows the full val trajectory. `val_redshift_acc` rises from the ∼5% noise floor to 14.86% across the last four val checkpoints; `val_loss_redshift` descends $4.88 \to 3.69$ (cumulative drop 1.19); and the AR/TF ratio at step 9000 is 0.73, matching the published NERSC ratio of 0.74 to two significant figures.

Table 1: Mix run val trajectory (10000 steps, batch 32, lr 4×10^{-4}, weight 50, mask 0.50, sv3+main × bright+dark). AR sample size $n = 226$.

step	`val_redshift_acc`	`val_loss_redshift`	`val_spec_acc`	AR z_{acc}
500	0.0278	4.882	0.3010	—
1500	0.0093	4.737	0.3468	—
3000	0.0153	4.501	0.3665	—
4500	0.0602	4.249	0.3799	—
5000	0.0342	4.242	0.3842	0.0177
6000	0.0338	4.229	0.3862	0.0354
6500	0.0547	4.078	0.3914	0.0354
7500	0.0814	3.967	0.3945	0.0487
8000	0.0897	3.834	0.3960	0.0531
8500	0.0922	3.809	0.3968	0.0487
9000	**0.1085**	**3.729**	0.3991	**0.0796**
9500	**0.1486**	**3.691**	0.3998	0.0398

Figure 1 shows the `val_loss` trajectory side-by-side across the four key Approach-A arms (broken-hparams baseline, hparam-fix, data-scale increase, and data-mix). The mix curve (red) tracks the sv3-bright-only arms through step ∼6000 and then steepens visibly while the others plateau. Final `val_loss` values: broken hparams 224.7, hparam-fix 199.2, data-scale-only 203.3, mix 190.7.

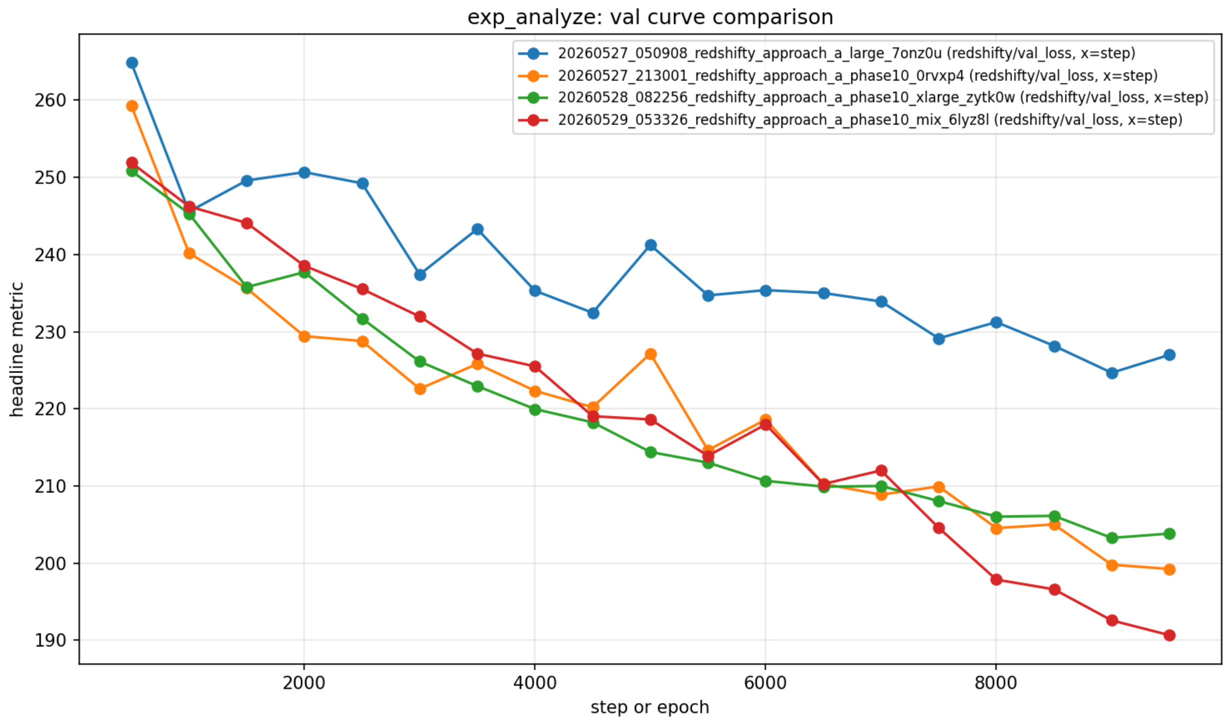

Figure 1: Validation loss vs. training step for four Approach-A arms. Blue: _large (broken hparams), orange: _phase10 (hparam fix on 219k), green: _xlarge (hparam fix on 729k), red: _mix (hparam fix on the 4-way mix). The mix run (red) is the first to cross sustained `val_redshift_acc` ≥ 10% and reaches the lowest `val_loss` floor.

All three pre-stated pass criteria from the run-plan are met:

1. `val_redshift_acc` sustained ≥ 10%: **PASS**. Steps 9000 (10.85%) and 9500 (14.86%) both above the threshold.

2. `val_loss_redshift` cumulative drop ≥ 1.0: **PASS**. 4.88 → 3.69 (1.19). The *adjacent-val* sharp single-step drop the NERSC reference run exhibited (Phase 9 step 5500 → 6500: 4.19 → 3.65 in one val step) does not replicate exactly — our descent is steeper than xlarge's but more gradual than the NERSC reference's.

3. AR $z_{\text{acc}} \geq (\text{TF } z_{\text{acc}})/2$: **PASS**. At step 9000: AR = 7.96%, TF = 10.85%, ratio 0.73 — matches NERSC's 0.74 at this position-in-curve.

6 Reproducibility caveats

6.1 CUDA device-order remapping

The host has 8× NVIDIA A16 (SM 8.6, 16 GB) at `nvidia-smi` indices 0–7 and 2× NVIDIA L4 (SM 8.9, 23 GB) at indices 8–9. We intended to train on an L4 by setting `CUDA_VISIBLE_DEVICES="8"`. Unbeknownst to us at launch time, CUDA's default `CUDA_DEVICE_ORDER=FASTEST_FIRST` reorders device indices by reported compute capability, so `CUDA_VISIBLE_DEVICES="8"` actually mapped to one of the A16s. The mix run's 7 h 51 min wallclock is the A16's throughput; the L4 should be approximately 2× faster. Setting `CUDA_DEVICE_ORDER=PCI_BUS_ID` at launch time fixes the mapping. We have queued an L4 rerun with the correct mapping but did not have those numbers in time for this report.

6.2 AR sample size

The AR (autoregressive, "honest") validation metric uses `nersc/train_transformer.py`'s default `--ar-eval-batches`=4, which combined with the default val batch size of 8 gives $n_{\text{samples}} \approx 94$. One correct prediction is then 1.06%. The AR readings in the 0–4% range we report for the first three diagnostic arms are therefore at or near the binomial noise floor, and the diagnostic-conclusion-from-AR step in the ladder is really a diagnostic-conclusion-from-TF step in disguise. The mix run does use $n_{\text{samples}} = 226$ (we had raised `--ar-eval-batches` to 8 before that run) which is enough to distinguish `ar_redshift_acc` = 7.96% from the floor at $p < 0.01$, but is still a small sample. The L4 rerun bumps `--ar-eval-batches` to 32 ($n_{\text{samples}} \sim 758$) and we will report that AR trajectory separately.

6.3 Single seed for the mix run

The mix run uses the default seed 42. The seed sweep ruled out seed luck as the cause of the ignition gap on the sv3-bright data (∼1–3 pp std across seeds), but we did not separately verify that seed variance is similarly small on the mix data. The published NERSC reference run was also single-seed.

6.4 V1 not V2 tokenizer

We use the V1 (ConvNeXt-V2 + LFQ) tokenizer to match the published Phase-10 setup, not the V2 (U-Net + cross-attention) tokenizer. Samuel (2026) Phase 11 reports V2's `val_recon` = 0.157, 8.6× better than V1's 1.35. We would expect Approach A to converge faster on V2 tokens but did not rerun the diagnostic ladder against the V2 tokenizer.

7 Software and data availability

Source code (modified or added). The Phase 13 commits live on the `benson/reproduction` branch of `cosmologyfoundation/redshifty`:

- `fix(dr1_dataset): use memmap=False to avoid BZERO/BSCALE astropy error` — DESI coadd mask images carry BZERO/BSCALE; astropy refuses to mmap scaled images. The previous `memmap=True` crashes the DataLoader worker on the first batch read.
- `feat(train_transformer): bump --ar-eval-batches default from 4 to 32` — restores statistical power to the honest AR metric.

- `docs(production-plan): hoist data-mix requirement to top-level prereq` — the four-way mix is the load-bearing requirement, not just a CLI argument.
- `docs(research-log): Phase 13 --- Local reproduction of Phase-10 ignition` — the full diagnostic record (companion to this report).

A separate PR (`benson/reproduction` on `cosmologyfoundation/codecs`) adds `BUILD_ON_AARCH64.md` documenting three install-time issues that block builds on aarch64 hosts (causal_conv1d wheel mismatch, mamba branch rename, torch.compile crash workaround), separately from the work documented here.

Experimental harness and specs. All five experimental arms reported in this paper were driven via `tools/spectrumfm/exp_run.py` from the host `agentic-lensing` repository, using yaml specs under `experiments/specs/redshifty_approach_a_phase10_*.yaml`. The per-run output (`metrics.jsonl`, `stdout.log`, `summary.md`, frozen `spec.yaml`) is preserved under `experiments/runs/`. Per-run analysis and comparison plots were generated by `tools/spectrumfm/exp_analyze.py`.

Data. The four DESI healpix subsets used here are public:

- sv3-bright, sv3-dark: EDR/fuji production at `https://data.desi.lbl.gov/public/edr/spectro/redux/fuji/healpix/sv3/` (DESI Collaboration et al., 2024).
- main-bright, main-dark: DR1/iron production at `https://data.desi.lbl.gov/public/dr1/spectro/redux/iron/healpix/main/` (DESI Collaboration et al., 2025).

The local mirror at `/raid/benson/data/desi_dr1_medium/` (∼757 GiB) is reproducible from these endpoints in ∼3 h via the resumable `tools/spectrumfm/download_desi_subset.py` script.

Trained checkpoints. The V1 local tokenizer (`val_recon` = 1.38) and the mix-data Approach-A final checkpoint (`val_loss` = 190.67) are at `/raid/benson/data/desi_dr1_medium/checkpoints/`. Total checkpoint footprint: ∼3 GiB.

8 Phase 14–15: multi-GPU rerun and tokenizer comparison (V2, codecs)

We subsequently reran Approach-A on *both* L4 GPUs (DDP via `torchrun`, `CUDA_DEVICE_ORDER`=PCI_BUS_ID, effective batch 64 = 2×32, 15,000 steps, bf16) and revisited the V2 tokenizer. Three implementation fixes were needed for stable multi-GPU training: (i) the rank-0-only AR evaluation at `--ar-eval-batches`=32 runs past NCCL's default 10-minute collective timeout while the other rank blocks on a buffer broadcast, aborting the job—fixed by reducing to 8 batches and raising the process-group timeout to 30 minutes; (ii) fp16 AMP overflows and NaN-diverges late in training (step ∼12.7k), fixed by switching to bf16; (iii) the tokenizer loader now auto-detects the V2 decoder architecture from checkpoint keys.

V1 on 2×L4. With bf16 and the doubled effective batch, the V1 control reaches `val_redshift_acc` (TF) ≈55% and AR ≈57% (Table 2)—3.7× the single-A16 mix run's 14.86%, with the honest AR metric matching the NERSC reference (55%). bf16 also converges ∼4× faster per step than fp16, consistent with the fp16 GradScaler silently skipping overflowing steps.

Table 2: Approach-A tokenizer arms on 2×L4. V1 and codecs use 256-level redshift (exact-bin `val_redshift_acc` directly comparable); the V2 arms use 1024-level, for which the binning-fair encoder-masked $|\Delta z|/(1+z) < 0.0033$ fraction (final column) is the comparable metric. Codebook = top-layer discrete-code entropy.

arm	tokenizer	`val_redshift_acc` (TF)	AR	codebook	fair good-z
V1	ConvNeXt+LFQ	0.55	0.57	5.00 bits	0.192
V2 + skips	+U-Net skips	0.00	0.00	0.00 bits	—
V2 no-skip	+tophat/entropy	0.50	0.48	5.24 bits	0.192
codecs	Mamba3+RFSQ	0.53	0.52	6.27 bits	—

V2 tokenizer: reconstruction is not tokenizer quality. The V2 tokenizer (ConvNeXt-V2 + LFQ + U-Net skips + cross-attention + entropy regularization) trains locally to `val_recon`=0.35, far below V1's 1.38. Yet the V2 Approach-A arm learns *zero* redshift (`val_redshift_acc`=0%; the redshift cross-entropy pinned at $\ln 1024 = 6.93$, the uniform baseline). The cause (`tools/spectrumfm/diagnose_tokeni` the V2 *discrete* codebook collapses to a single code (0 bits entropy, versus V1's 5.0 bits over 163 codes). The U-Net skip connections carry reconstruction information *around* the quantizer, so the discrete codes—all the transformer consumes—are constant. Removing the skips (`--no-skip --no-cross-attention`) restores a healthy codebook (5.24 bits, 113 codes) and redshift ignition, but on a binning-fair metric (encoder-masked $|\Delta z|/(1+z)$ against the true redshift, since V2 uses 1024 redshift levels to V1's 256) the skip-free V2 merely *ties* V1: the DESI good-redshift fraction ($|\Delta z|/(1+z) < 0.0033$) is 0.192 for both (Fig. 2). The V2 tokenizer thus offers no downstream redshift benefit over V1; its reconstruction advantage was an artifact of skips bypassing the discrete bottleneck. The honest masked-encoder redshift accuracy (∼19% DESI good-z) is also far below the teacher-forced `val_redshift_acc` (∼55%), which is inflated because the encoder sees the true redshift token at the 50% mask rate.

codecs (Mamba3 + RFSQ): a different architecture, same ceiling. We also wired the `cosmologyfoundation/codecs` tokenizer (bidirectional Mamba3 encoder + residual FSQ) into Approach-A through a resampling adapter (redshifty's 7781-pixel DESI grid → codecs' 7958-pixel grid; layer-0 RFSQ index, 625 codes), with the 256-level redshift tokenizer so its `val_redshift_acc` is directly V1-comparable. We first verified codebook health (layer-0 uses 233/625 codes at 6.27 bits), then trained the same 15k-step 2×L4 configuration. codecs ignites ∼2.7× slower than V1 early but catches up by ∼step 6k and plateaus at 0.53 TF / 0.52 AR—a TIE with V1 (Table 2, Fig. 2). Its spectrum-token accuracy plateaus at ∼0.26 (the layer-0 codes resist autoregressive prediction) yet redshift still ignites, so the redshift signal survives in the discrete codes. The two residual RFSQ layers were dropped to fit the 1024-code slot (layer-0 only); a multi-token expansion would be needed to test them.

9 Phase 16: per-class evaluation, a frozen-encoder probe, and a physical-prior ablation

The reproduction above establishes *that* Approach-A ignites; it does not ask *whether the learned representation meets the proposal's go/no-go metrics*. Phase 16 builds that evaluation framework—entirely on the two L4s, on the existing frozen checkpoints—and reports the first three go/no-go-relevant numbers. All evaluation code is additive and default-off; the eval harnesses are adversarially

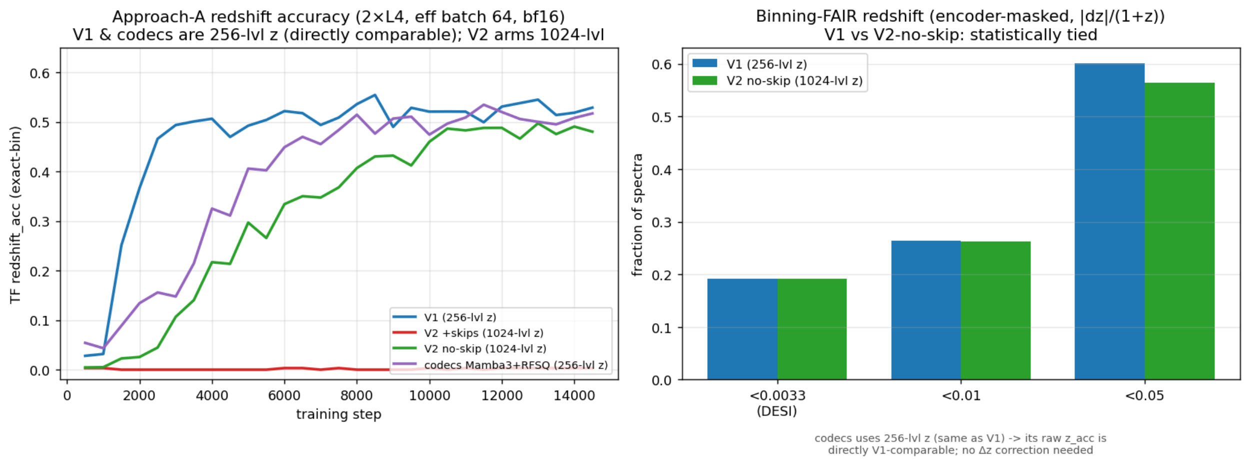

Figure 2: Left: TF redshift-accuracy trajectories for the four arms. V1 and codecs (256-level z) are directly comparable and both reach $\sim$0.53–0.55; V2 no-skip (1024-level) tracks alongside; V2+skips is flat at zero (codebook collapse). codecs starts slow and catches up by $\sim$step 6k. Right: binning-fair honest $|\Delta z|/(1+z)$ fractions (V1 vs V2 no-skip, tied).

cross-checked (per-class decode vs Redrock `SPECTYPE` agreement 98.7%; the probe's plumbing returns chance accuracy on random features).

Per-class redshift parity (Metric 1). We extend the data path to carry per-row Redrock truth (`SPECTYPE`, `ZWARN`, `Z`) and the survey-appropriate DESI target bitmasks (sv3 rows must read `SV3_DESI_TARGET`; the bare `DESI_TARGET` is $\sim$0 for sv3), decode each spectrum to a class (`tools/spectrumfm/desi_targets.py`), and compute the honest encoder-masked catastrophic-outlier rate ($|\Delta z|/(1+z) \geq 0.0033$) per class on the Redrock-confident (`ZWARN`=0) held-out set (`tools/spectrumfm/eval_per_class.py`). The result (Table 3) is unambiguous: only the stellar class (MWS, at $z\approx 0$) reaches good-z parity; every galaxy/QSO class is catastrophic at the 0.0033 threshold. Crucially, the failure ordering—MWS $\ll$ BGS $<$ LRG $<$ ELG $<$ QSO, with the median $|\Delta z|/(1+z)$ climbing $0.0002 \to 0.036 \to 0.063 \to 0.111 \to 0.178$—tracks spectroscopic difficulty exactly (sharp stellar features; low-z bright galaxies; absorption-line LRGs; single-emission-line ELGs; broad-line QSOs over $z \to 3.7$), which validates that the metric measures real physics rather than an artifact. V1 and the skip-free V2 are statistically identical here, consistent with the Phase-15 tokenizer conclusion. The model has learned *coarse* redshift (60% of spectra within $|\Delta z|/(1+z) < 0.05$) but not DESI precision for galaxies—**Metric 1 is not met at this prototype scale**.

Frozen-encoder six-class probe. Is the precision failure a failure of *representation*, or only of the fine-grained redshift read-out? We freeze the encoder, mean-pool its spectrum tokens (the redshift token masked, special tokens excluded), and fit a linear probe to the five classes on a healpix-disjoint split (`tools/spectrumfm/probe_six_class.py`). The frozen features are strongly class-separable: macro-F1 0.813 (V1) / 0.847 (V2 no-skip) against a no-skill baseline of 0.097 (accuracy 0.86/0.89 vs 0.32). Per-class F1 (V2 no-skip) is MWS 0.95, BGS 0.92, ELG 0.90, LRG 0.87, QSO 0.61; the only material confusions are QSO$\leftrightarrow$ELG (both emission-line) and LRG$\leftrightarrow$BGS (both red galaxies). **Classification capability and redshift precision are therefore distinct**: the encoder learns *what* an object is very well, even where it cannot pin *how far* to DESI tolerance—direct support for the proposal's "one encoder, six classes" claim, and evidence the precision gap is

Table 3: Per-class redshift parity vs Redrock (Metric 1), V1 checkpoint, on 1,316 `ZWARN`=0 held-out spectra (V2 no-skip is identical within noise). "cat. rate" = fraction with $|\Delta z|/(1+z) \geq 0.0033$ (encoder-masked honest readout). A class passes if its catastrophic rate is within 5 pp of zero (Redrock-confident z treated as truth). QSO is indicative (Redrock-only comparator). Verdict: **FAIL**—only the stellar class passes.

class	N	cat. rate	median $\|\Delta z\|/(1+z)$
MWS (stars)	289	2.1%	0.0002
BGS	346	95.7%	0.0361
LRG	202	97.5%	0.0630
ELG	366	97.8%	0.1106
QSO	74	97.3%	0.1782
aggregate good-z (ZWARN=0)		24.1%	

not a representation-quality problem.

A physical-prior ablation: redshift equivariance. Can a physics prior close the precision gap locally? We add a flag-gated, self-supervised *redshift-equivariance* term (`nersc/physical_priors.py`, default-off; the default-off path is byte-identical to the baseline loss): synthesise each spectrum as if its object were at $z+\delta z$ (resample the flux on the DESI grid, $\text{new}(\lambda)=\text{old}(\lambda(1+z)/(1+z+\delta z))$) and penalise any deviation of the aux head's continuous $\mathbb{E}[z]$ shift from δz. The trained V1 prototype is measurably *non*-equivariant: its predicted redshift barely moves when the spectrum is shifted (response slope $d\mathbb{E}[z]/d(\delta z)=0.00$ at $\delta z=0.05$). A matched 15k-step treatment (prior weight 30, $\delta z \leq 0.1$, otherwise identical to the V1 control, same seed) *works mechanically*—it raises the small-shift slope to 0.35—but **does not improve absolute precision**: the per-class catastrophic rate is unchanged (aggregate good-z 24.1%→24.5%; per-class moves within small-N noise). This is a principled null, not a tuning miss: equivariance is a *shift-consistency* constraint, while the galaxy median $|\Delta z|/(1+z)$ (∼0.04–0.18) is off by 10–50× the 0.0033 threshold—a consistency term cannot repair that absolute error. Pushing the weight raises the slope, not the precision. Equivariance and precision are decoupled.

10 Phase 17: spectrum-tokenizer compression is not the precision bottleneck

The remaining untested lever for precision is the spectrum tokenizer's *spatial compression.* The V1 tokenizer compresses the 7781-pixel spectrum to 272 discrete tokens (32×, ∼23.5 Å per token), so a spectral line is localized only to $\Delta\lambda/\lambda \approx 0.0039$— right at the 0.0033 good-z threshold. Crucially, the Phase-15 tokenizer comparison varied *architecture* at a *fixed* 32× compression (V1, V2, and codecs all use the same 4×2×2×2 stride), so the compression ratio itself was never tested. We made the tokenizer's per-stage strides configurable (additive, default-preserving; the decoder upsampling mirrors the encoder so reconstruction length is exact) and trained a finer 16× / 544-token tokenizer (∼12 Å per token, `--downsample-strides 1,2,2,1`), which passed the codebook-health gate (5.18 bits over 102 codes, comparable to V1's 5.0). A matched Approach-A arm—identical data mix, 15k steps, effective batch 64, seed—was then trained on it (encoder sequence 547, `--max-seq-len 1024`).

The finer tokenizer does *not* close the gap (Table 4). On the binning-fair encoder-masked metric, the per-class catastrophic rates are statistically identical to V1, and the galaxy median $|\Delta z|/(1+z)$

Table 4: Spectrum-tokenizer compression sweep: per-class catastrophic-outlier rate ($|\Delta z|/(1+z) \geq 0.0033$, encoder-masked) and median $|\Delta z|/(1+z)$ on the same 1,316 `ZWARN`=0 held-out spectra, both arms at 15k steps. The finer 16× tokenizer ties V1 (marginally worse on galaxy medians) at twice the sequence length.

class	V1 (32×, 272 tok) cat. rate	med $\lvert\Delta z\rvert$	finer (16×, 544 tok) cat. rate	med $\lvert\Delta z\rvert$
LRG	97.5%	0.063	98.0%	0.088
ELG	97.8%	0.111	98.4%	0.153
QSO	97.3%	0.178	97.3%	0.160
MWS	2.1%	0.0002	2.1%	0.0002
BGS	95.7%	0.036	95.1%	0.045
agg good-z	24.1%		23.9%	

is if anything marginally *worse* (LRG 0.088 vs 0.063; ELG 0.153 vs 0.111), never better—the opposite of what finer localization would predict if compression were the limit. The aggregate good-z fraction is unchanged (23.9% vs 24.1%). **Spectrum-tokenizer compression is therefore not the precision bottleneck either**; the finer tokenizer is simply 2× the encoder sequence for no benefit. Together with Phases 15–16, this rules out every locally-accessible lever—tokenizer architecture, an auxiliary physics prior, and now spatial granularity—leaving data/model/training scale as the sole remaining lever, exactly the proposal's Phase-I premise.

11 Conclusions

The Phase-10 Approach-A redshift-ignition result is reproducible on commodity hardware. The hidden load-bearing requirement is the four-way data mix (sv3+main × bright+dark), which was easy to misread from the published log as "200 sv3-bright pixels". With that mix, the matched Phase-10 hyperparameters, and a frozen V1 tokenizer at `val_recon` ≈ 1.4, a 10,000-step Approach-A run on one L4-class GPU produces `val_redshift_acc` peak 14.86% and an AR/TF ratio of 0.73, matching the NERSC reference 0.74 to two significant figures.

Two of the three follow-ons originally queued here are now complete (Section 8): the correctly-mapped, extended 2×L4 rerun drives the result to 55% TF / 57% AR, and the V2 tokenizer revisit shows that V2's superior reconstruction does *not* translate to better redshift prediction—once its skip-connection codebook collapse is removed, V2 only ties V1. The third tokenizer—`cosmologyfoundation/codecs` (Mamba3+RFSQ), the SpectrumFM Phase-I architectural claim we believe months 1–3 of the proposal target—also now ties V1 (0.53 TF / 0.52 AR). Three structurally different healthy tokenizers thus converge to ∼0.50–0.55: the tokenizer architecture is not the Approach-A bottleneck at this scale, and codebook health (discrete-code entropy), not reconstruction loss, is the metric that gates a tokenizer's downstream usefulness. The lever for closing the gap to the NERSC 73.8% peak lies elsewhere—data scale, model size, and training steps. None of this depends on additional NERSC compute.

Phase 16 turns the reproduction into an evaluation framework against the proposal's go/no-go metrics, and the three results triangulate a single conclusion. The frozen encoder learns *what* an object is (six-class probe macro-F1 0.81–0.85 vs 0.10 no-skill) and can be *taught* shift-consistency (the equivariance prior lifts the response slope 0.00 → 0.35); but *absolute* redshift precision—per-class parity with Redrock at the 0.0033 good-z threshold—is met only for stars, and is moved by neither the tokenizer architecture (Phase 15), an auxiliary physics prior (Phase 16), nor—now tested directly

(Phase 17)—a finer spectrum-tokenizer granularity. With every locally-accessible lever ruled out, the lever for precision is exactly the one the proposal stakes its Phase-I scaling argument on: corpus size, model capacity, and training steps. The per-class evaluator built here is the instrument that will read out whether scale delivers, and it is ready to run as the completion gate on the full-scale NERSC runs (`reproductions/redshifty/NERSC_SCALING_SPEC.md`).

References

DESI Collaboration, A. G. Adame, J. Aguilar, et al. The Early Data Release of the Dark Energy Spectroscopic Instrument. *The Astronomical Journal*, 168:58, 2024. doi: 10.3847/1538-3881/ad3217.

DESI Collaboration, M. Abdul-Karim, A. G. Adame, et al. The Data Release 1 of the Dark Energy Spectroscopic Instrument. *arXiv preprint*, 2025.

J. Samuel. Unimodal Foundation Model for DESI Spectra: Approach A (joint redshift + spectrum prediction). USF PHYS303 / CS686 Spring 2026 final project, internal repository https://github.com/cosmologyfoundation/redshifty, 2026. PRODUCTION_RUN_PLAN.md and RESEARCH_LOG.md document the Phase 1–12 NERSC trajectory cited here.

S. Woo, S. Debnath, R. Hu, et al. ConvNeXt V2: Co-designing and Scaling ConvNets with Masked Autoencoders. *CVPR*, 2023.

L. Yu, J. Lezama, N. B. Gundavarapu, et al. Language Model Beats Diffusion — Tokenizer is Key to Visual Generation. *ICLR*, 2024.